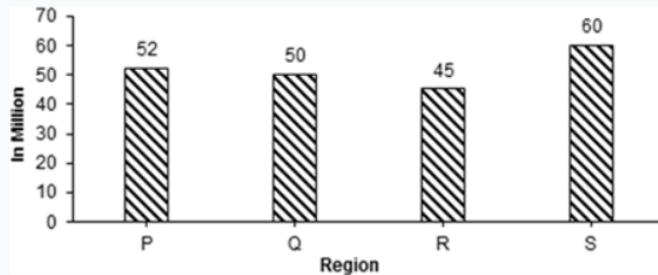


DI – BAR GRAPHS

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The details of the National Readership Survey (NRS) report regarding the readership (in million) of four major newspapers in regions P, Q, R and S are as follows:



The following table gives the percentage of reader's for the newspapers in the four regions.

Region \ News Papers	P	Q	R	S
A	30%	25%	20%	30%
B	40%	25%	30%	30%
C	10%	30%	40%	20%
D	20%	20%	10%	20%

How many (in million) read Newspaper B in region P?

☐ a) 20.8 (Correct Answer - ✓)

☐ b) 30

☐ c) 16.5

☐ d) 22.5

Solution:

In region P, the total number of readers = 52 million.

The number of readers reading newspaper B

$$= \frac{40}{100} \times 52 \text{ million}$$

$$= \frac{2}{5} \times 52 \text{ million}$$

$$= 20.8 \text{ million.}$$

Choice (A)

What is the ratio of the number of readers of newspaper A in region Q to the number of readers of newspaper B in region S?

☐ a) 11 : 12

☒ b) 25 : 36 (Correct Answer - ✓)

☐ c) 18 : 17

☐ d) None of these

Solution:

The number of readers of newspaper A in region Q = $\frac{25}{100} \times 50 \rightarrow (1)$

The number of readers of newspaper B in region S = $\frac{30}{100} \times 60 \rightarrow (2)$

The required ratio = $25 \times 50 : 30 \times 60 = 25 : 36$

Choice (B)

What is the total number (in million) of readers of newspaper C in the four regions together?

☐ a) 26

☐ b) 52.5

☐ c) 48.5

☒ d) 50.2 (Correct Answer - ✓)

Solution:

The required number of readers

$$\begin{aligned} &= \frac{10}{100} \times 52 + \frac{30}{100} \times 50 + \frac{40}{100} \times 45 + \frac{20}{100} \times 60 \\ &= 5.2 + 15 + 18 + 12 = 50.2 \text{ million.} \end{aligned}$$

Choice (D)

What is the ratio of the total readers of newspaper B to that of newspaper C?

☐ a) 341 : 171

☒ b) 324 : 251 (Correct Answer - ✓)

☐ c) 3 : 5

☐ d) None of these

Solution:

The total readers of B

$$= \frac{40}{100} \times 52 + \frac{25}{100} \times 50 + \frac{30}{100} \times 45 + \frac{30}{100} \times 100$$

$$= 20.8 + 12.5 + 13.5 + 18 = 64.8$$

The total readers of C = 50.2

The required ratio = $64.8 : 50.2 = 324 : 251$.

Choice (B)

In which region is the ratio of the readers of newspaper A to that of newspaper B, the least?

☐ a) R (Correct Answer - ✓)

☐ b) P

☐ c) Q

☐ d) Cannot be determined

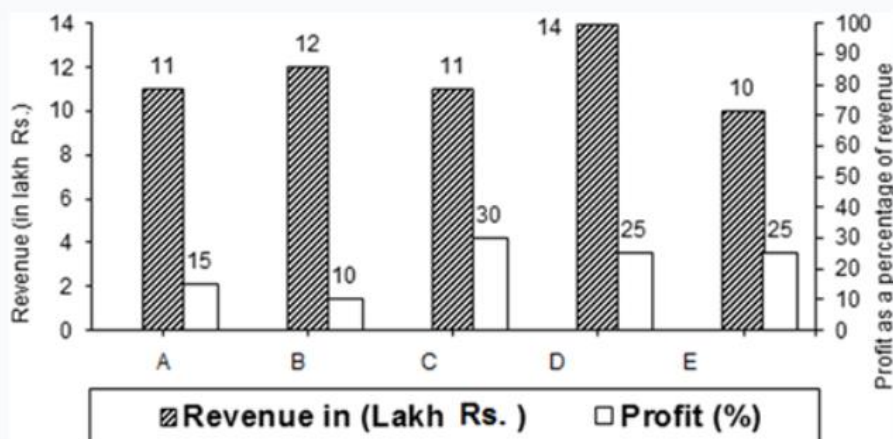
Solution:

In region R, the ratio of the readers of A to that of B is the lowest and it is 2 : 3.

Choice (A)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The bar graph provides the revenue and the profit as a percentage of the revenue for five companies.



For which company is the expenditure, the least?

☐ a) B

☒ b) E (Correct Answer - ✓)

☐ c) A

☐ d) C

Solution:

	A	B	C	D	E
Revenue (Lakhs)	11	12	11	14	10
Profit (Lakhs)	1.65	1.2	3.3	3.5	2.5
Expenditure (Lakhs)	9.35	10.8	7.7	10.5	7.5

From the table it is clear that the expenditure is least for E.

Choice (B)

If the profit percentage is calculated on the expenditure, which company has the highest profit percentage?

☒ a) C (Correct Answer - ✓)

☐ b) D

☐ c) E

☐ d) B

Solution:

	A	B	C	D	E
	$\frac{1.65}{9.35}$	$\frac{1.2}{10.8}$	$\frac{3.3}{7.7}$	$\frac{3.5}{10.5}$	$\frac{2.5}{7.5}$
Percentage	0.17	0.11	0.42	0.33	0.33

∴ The profit percentage is highest for company C.

Choice (A)

What is the average profit (in lakh Rs.) of all the five companies?

- ☐ a) 1.8
- ☐ b) 2.65
- ☐ c) 3.42

☒ d) 2.43 (Correct Answer - ✓)

Solution:

$$\begin{aligned}\text{The average profit (in lakh ₹.)} &= 1.65 + 1.2 + 3.3 + 3.5 + 2.5. \\ &= \frac{12.15}{5} = 2.43.\end{aligned}$$

Choice (D)

The expenditure of B is what percentage more/less than that of C?

☒ a) 40.25% (Correct Answer - ✓)

- ☐ b) 30%
- ☐ c) 42.75%
- ☐ d) 18%

Solution:

$$\text{The required percentage} = \frac{10.8 - 7.7}{7.7} \times 100 = 40.25\%$$

Choice (A)

What is the aggregate expenditure (in lakh Rs.) of the five companies?

- ☐ a) 29.35
- ☐ b) 18.75

☒ c) 45.85 (Correct Answer - ✓)

- ☐ d) 46.25

Solution:

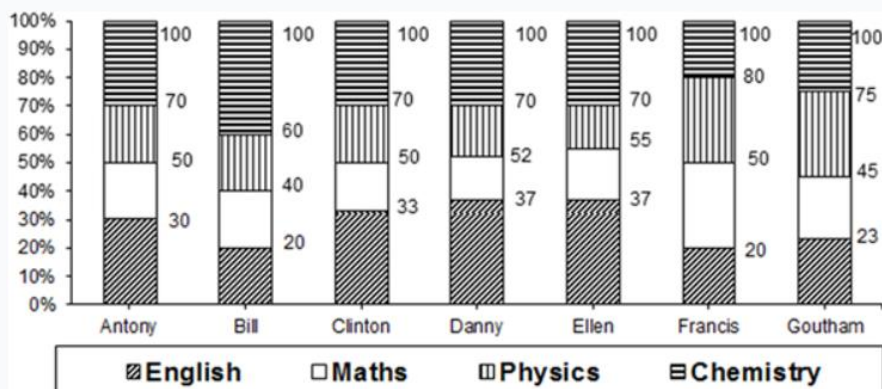
$$\text{The total expenditure} = ₹.45.85 \text{ lakhs}$$

Choice (C)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The graph shows the percentage break up of marks scored by Antony, Bill, Clinton, Dancy, Ellen, Francis and Goutham in four subjects in their class XII exam.

The total marks scored by the above seven students are 370, 280, 380, 365, 375, 360 and 390 respectively. The maximum marks in each subject is 150.



Who scored the highest marks in Maths?

☐ a) Goutham

☒ b) Francis (Correct Answer - ✓)

☐ c) Danny

☐ d) Anthony

Solution:

$$\text{The marks scored by Antony} = \frac{20}{100} \times 370 = 74.$$

$$\text{The marks scored by Bill} = \frac{20}{100} \times 280 = 56$$

$$\text{The marks scored by Clinton} = \frac{17}{100} \times 380 = 64.6$$

$$\text{The marks scored by Danny} = \frac{15}{100} \times 365 = 54.75$$

$$\text{The marks scored by Ellen} = \frac{18}{100} \times 375 = 67.5$$

$$\text{The marks scored by Francis} = \frac{30}{100} \times 360 = 108.$$

$$\text{The marks Goutham scored} = \frac{22}{100} \times 390 = 85.8.$$

Francis scored the highest in Maths.

Choice (B)

The student whose marks in Chemistry is closest to the average marks scored by all the seven students in that subject is _____.

☐ a) Clinton

☒ b) Danny (Correct Answer - ✓)

☐ c) Antony

☐ d) Goutham

Solution:

The average score in Chemistry

$$\begin{aligned} &= \frac{1}{7} \left[\frac{30}{100} \times 370 + \frac{40}{100} \times 280 + \frac{30}{100} \times 380 + \frac{30}{100} \times 365 \right. \\ &\quad \left. + \frac{30}{100} \times 375 + \frac{20}{100} \times 360 + \frac{25}{100} \times 390 \right] \\ &= \frac{111 + 112 + 114 + 109.5 + 112.5 + 72 + 97.5}{7} = \frac{728.5}{7} = 104. \end{aligned}$$

∴ Danny's score in Chemistry is the closest to the average score.

Choice (B)

What is the ratio of the marks scored by Clinton in Physics to those scored by Bill in English?

☒ a) 19 : 14 (Correct Answer - ✓)

☐ b) 5 : 7

☐ c) 6 : 5

☐ d) None of these

Solution:

The marks scored by Clinton in Physics = 76.

The marks scored by Bill in English = $\frac{20}{100} \times 280 = 56$.

The required ratio 76 : 56
= 19 : 14.

Choice (A)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

In English, the marks scored by Ellen are approximately what percentage more than those by Goutham?

Solution:

$$\text{The marks scored by Ellen in English} = \frac{37}{100} \times 375 = 138.75.$$

$$\text{The marks scored by Goutham in English} = \frac{23}{100} \times 390 = 89.7.$$

$$\text{The required percentage} = \frac{138.75 - 89.7}{89.7} \times 100 = 54.6\%$$

Ans : (55)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

What is the total marks scored by all the seven students in Physics (approximated to the closest integer)?

Solution:

The total marks scored in Physics by all the students

$$\begin{aligned} & \frac{20}{100} \times 370 + \frac{20}{100} \times 280 + \frac{20}{100} \times 380 + \frac{18}{100} \times 365 + \frac{15}{100} \times 375 + \frac{30}{100} \times 360 + \frac{30}{100} \times 390 \\ &= 74 + 56 + 76 + 65.7 + 56.25 + 108 + 117 \\ &= 553. \end{aligned}$$

Ans : (553)